

NEPSE data

- Daily closing level and turnover
- 3 Jan 2010 – 4 Aug 2026
- Source: ShareSansar

Foreign market data

- 33 equity indices; 27 retained after a coverage screen
- 3 Jan 2010 – 5 Aug 2026
- Source: Yahoo Finance

Feature construction (all predictors lagged by at least one day)

NEPSE features (17)

- Lagged returns, momentum, volatility
- Distance from moving averages
- Lagged turnover

Foreign market features (135)

- Lagged returns (1–3 days)
- 5-day momentum and volatility
- For each of 27 markets

Regional features (18)

- Lagged regional average return
- 5-day regional momentum
- 9 regions

Calendar features (2)

- Day of week
- Month

Information sets (6)

1. NEPSE-only

2. Foreign-only

3. Combined

4. Global factors (PCA)

5. NEPSE + PCA

6. NEPSE + PLS

Models (8)

Classical machine learning (6)

- Ordinary least squares (OLS)
- Ridge regression
- Random Forest (RF)
- Extra Trees (ET)
- Gradient Boosting (GBR)
- XGBoost (XGB)

Recurrent neural networks (2)

- Long short-term memory (LSTM)
- Gated recurrent unit (GRU)
- 20-day input sequences

Training and evaluation

Chronological split

- Train: 6 Jan 2010 – 30 Apr 2023 (2,549 days)
- Test: 2 May 2023 – 4 Aug 2026 (731 days)

Model fitting

- Imputation and scaling fitted on training data only
- Ridge penalty chosen by time-series CV; fixed settings for other models
- Each model fitted once on the training period

Evaluation

- Hold-out RMSE, MAE, R^2 , directional accuracy
- Diebold–Mariano and Pesaran–Timmermann tests
- Walk-forward check (gradient boosting, 6 origins)